fact that this technology has only been around for less than 150 years is a testament to human ingenuity and engineering prowess. We've got the absolute best ICEs in the fastest cars but the problem still remains; Internal Combustion Engines are still not zero-emission vehicles. It's imperative that we as a species move towards a cleaner, greener mode of transport. And, that's what EVs are here for.

But, how do you convince a population that is used to the roar of an ICE to switch to an EV? For starters, car manufacturers should start marketing the fact that EVs, in every way, shape and form are much safer than the good old ICE. Think about it, EVs are essentially technological devices rather than a basic mechanical engine. There are fewer parts, fewer gears etc, making it much less susceptible to mechanical failure. In fact, the Insurance Institute for Highway Safety (IIHS) has stated that EVs rank among the highest when it comes to safety.

IIHS President David Harkey states, "It's fantastic to see more proof that these vehicles are as safe as or safer than gasoline- and diesel-powered cars. We can now say with confidence that making the U.S. fleet more environmentally friendly doesn't require any compromises in terms of safety."

EVS BUILT WITH SAFETY AS A PRIORITY

Even in 2022, the Tesla Model 3 crushed all its safety tests and came out with flying colours. It even took home the prize for the best roof protection in a sedan as well. And it's not just Tesla that's dominating safety tests, the Audi e-Tron, Volvo XC40 Recharge and Hyundai Ioniq 5 also tested incredibly well in crash and trauma tests.

Another incredibly important point to consider when comparing EV safety to an ICE is the presence of fuel! Substances like petrol and diesel are flammable and very volatile. Sure, batteries are also flammable but not to the extent of combustible fuel. In case of a severe accident or car crash, the chances of the vehicle catching fire through the exposure of fuel are much higher than a battery exploding. EVs have also been subjected to much stricter norms and scrutiny that every little flaw is viewed under a magnifying glass. Fair enough, it is a new technology and must be monitored for any life-threatening issues that could potentially pop up.

ENGINE OR NO ENGINE?

The fact that EVs do not come with massive front-mounted engines is also another feather in its cap when it comes to vehicle safety. What this allows car manufacturers to do is bolster the front end of the vehicle with added reinforcement. The front end of EVs are sturdier, tougher and can

take more punishment than a soda can with an ICE. Again, Tesla stands out in this field with the Model 3 and Model Y being the best of the bunch. The crumple zone on EVs is much sturdier and will definitely help prevent serious injury in a collision.

Another standard on EVs is that the centre of gravity is very low. The battery pack is sealed and mounted on the underside of the vehicle adding an extra bit of weight. This stabilises the EV preventing it from rolling over and causing even more damage. So, if you're ever involved in an accident and you're in an EV, you'll be relatively safer than in a normal ICE vehicle simply because you will not end up upside down. There are a ton of videos on Youtube which demonstrate exactly this. EVs are put through a meat grinder of tests and almost all the time, they'll end up straight and not rolled over.

A fear that most people have when it comes to EVs is being electrocuted when it is raining. Again, EVs are designed in such a way that the charging connections only allow electricity to flow when contact between the plug and the car is complete. Almost all EVs are well protected against moisture and humidity. These vehicles have been designed with user safety as the number one priority. EVs are also designed to be "intrinsically safe". What this suggests is that anytime a defect is detected, the flow of power is immediately shut off, preventing any further problems.If you do get into a bad accident, the battery will be disconnected from the other high-voltage components and cables within milliseconds, thus preventing an electrical fire. There is a 12V battery that will power up the headlights and such but nothing that will harm you.

Look, EVs do still catch on fire from time to time, the technology isn't perfect as yet. An issue called thermal runway takes place when battery cells set off a reaction that sets the battery on fire. But again, these incidents are few and far between and will most likely be phased out in future EVs. In fact, manufacturers such as Volvo have been working